Stable Diffusion Basics

Foreword

- AI has made remarkable breakthroughs in visual art creation. Among them, the AI painting model Stable Diffusion is particularly noteworthy. This model has powerful image generation capabilities and is profoundly changing approaches to art creation, design, and visual content generation.
- Despite its impressive potential in art creation, Stable Diffusion has many limitations. This course introduces the capabilities of Stable Diffusion to help you better understand and utilize it to integrate art and technology.

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Objectives

- Upon completion of this course, you will understand:
  - The basics of diffusion models
  - The structure of the Stable Diffusion model
  - The basic applications of the Stable Diffusion model

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1. **Diffusion Model Basics**
2. Stable Diffusion Model Structure
3. Stable Diffusion Basic Applications

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Diffusion Model Definition

- Diffusion models are generative models that perform well in image generation. They simulate a diffusion process that goes from order to disorder and then back to order, and generate data by gradually adding and then removing noise in the process.
- Diffusion models involve two main phases: forward diffusion and reverse diffusion (or denoising).
  - **Forward diffusion**: In this phase, a model gradually adds noise to the raw data until it becomes almost random noise. This process is usually predefined and follows a specified path.
  - **Reverse diffusion**: In this phase, a model gradually restores the original ordered data from the noise. During this process, the model learns how to effectively restore data that contains a large amount of noise into clean data. This phase usually requires a trained deep neural network.

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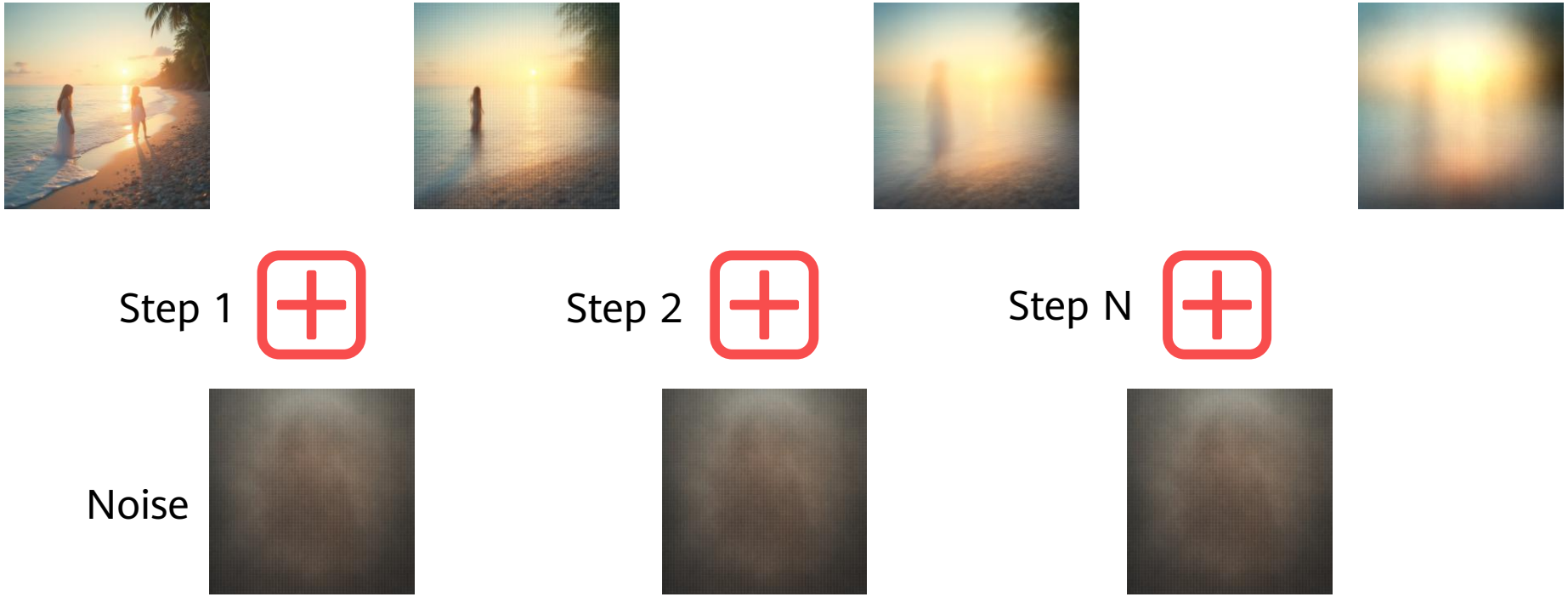
Diffusion Model Training Phase (1)

- In forward diffusion, the model gradually adds noise to data (such as an image) to convert it into a noise sample of Gaussian distribution. This process is usually fixed, and the amount of noise added in each step can be controlled by a predefined noise schedule.



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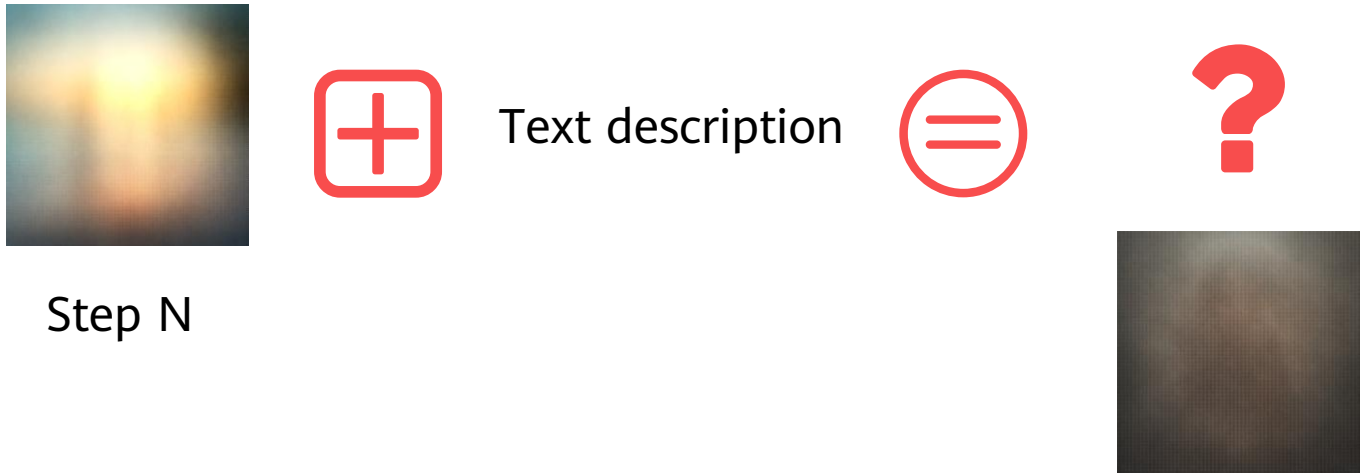
Diffusion Model Training Phase (2)

- In reverse diffusion, an image with noise is input to the model. The model predicts the noise added in each step.
- During training, the model's input also includes the text description of the image.



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Diffusion Model Inference

- In the inference phase, the model input is a random noise. It represents the random data state that complies with the Gaussian distribution. During inference, the model predicts and removes the noise. Predicting noise is easier than predicting a denoised image.

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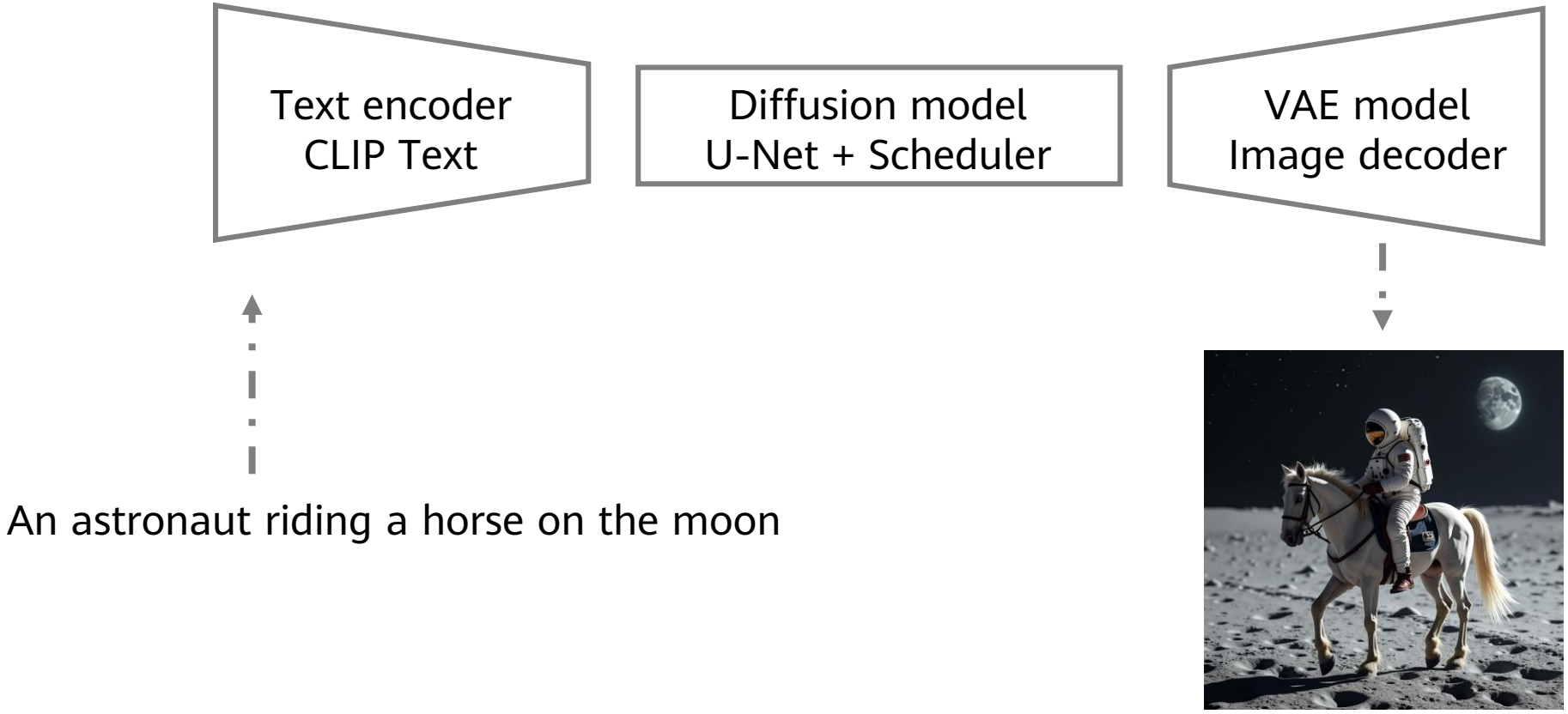
Introduction to Stable Diffusion

- Stable Diffusion is a text-conditioned latent diffusion model.



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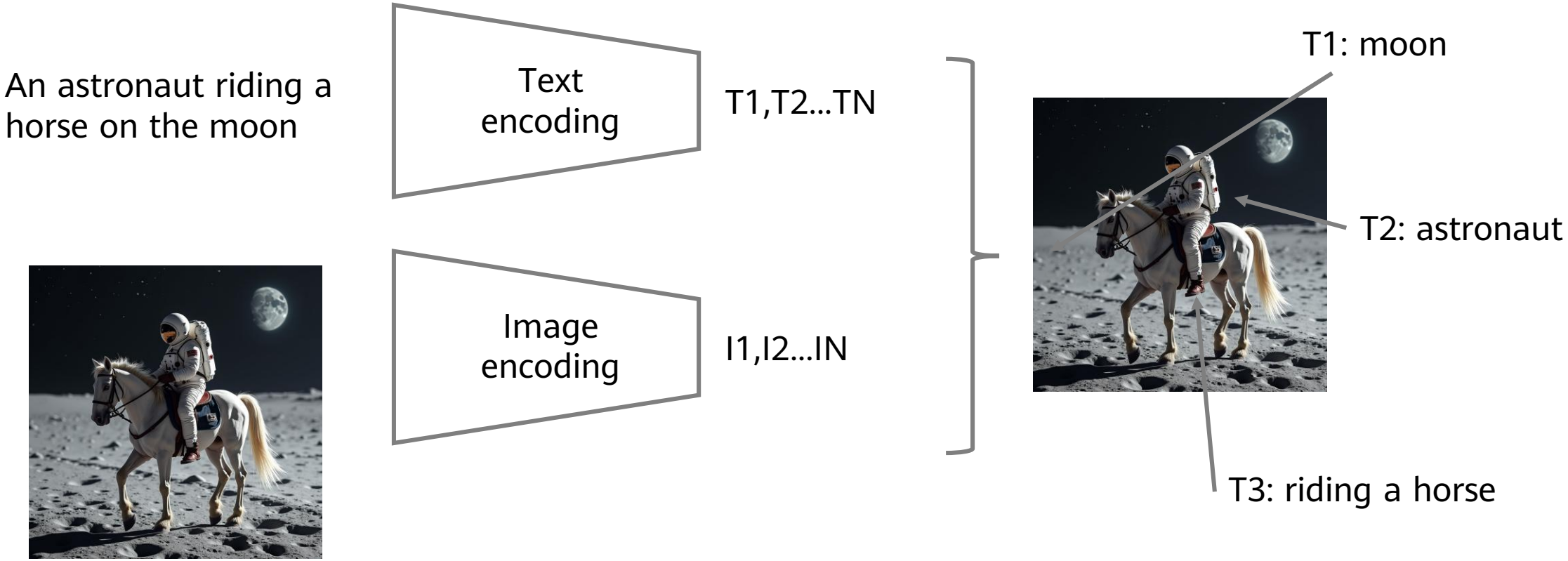
CLIP Text Encoder

- Stable Diffusion uses a pre-trained model called Contrastive Language-Image Pre-training (CLIP). The CLIP text encoder converts text into a feature vector that can be compared with the image's feature vector.



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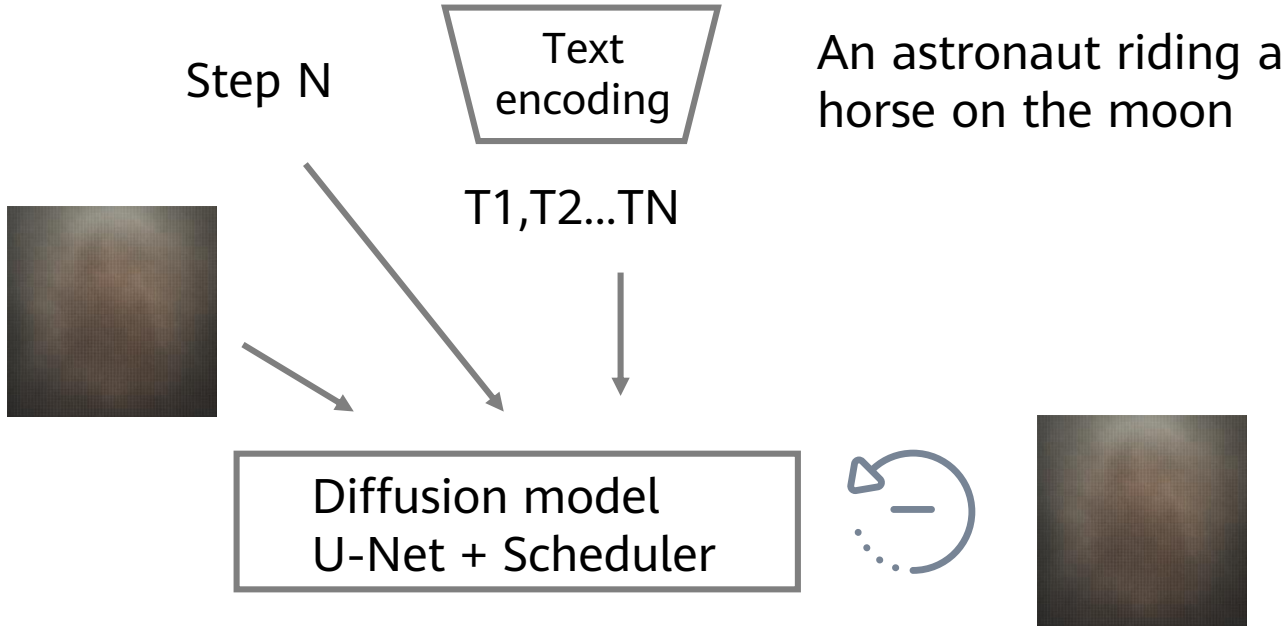
U-Net

- U-Net is the core of Stable Diffusion.
- It has three inputs:
  - A noisy image
  - A time tag
  - Other condition variables
- A noise output is obtained after layer-by-layer operations. The noise output can be used to denoise the image.



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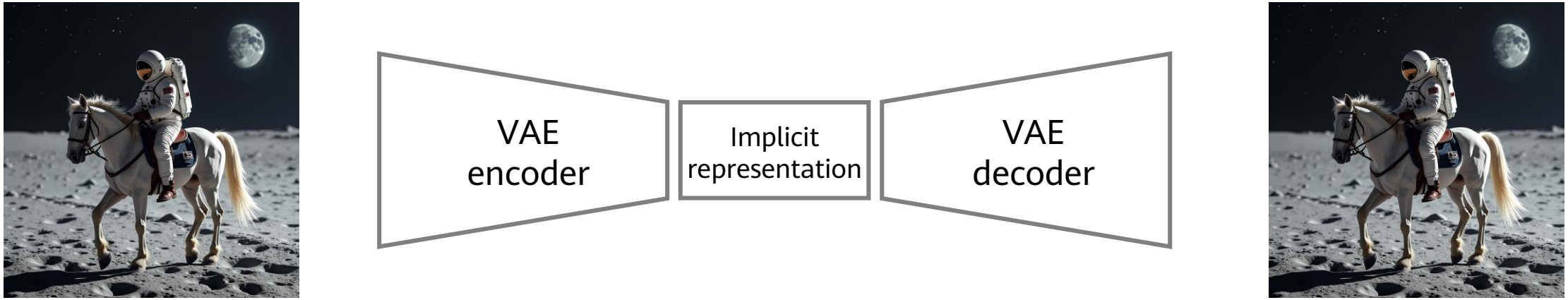
VAE Model

- A larger input image requires stronger computing power. This is because the number of operations increases quadratically with the input size.
- This problem can be resolved using the variational autoencoder (VAE), an independent model used to compress an image into a smaller space dimension.
- Images usually contain a large amount of redundant information. We can train a VAE using a large amount of image data to map an image to a smaller implicit representation and then map the representation to the original image.



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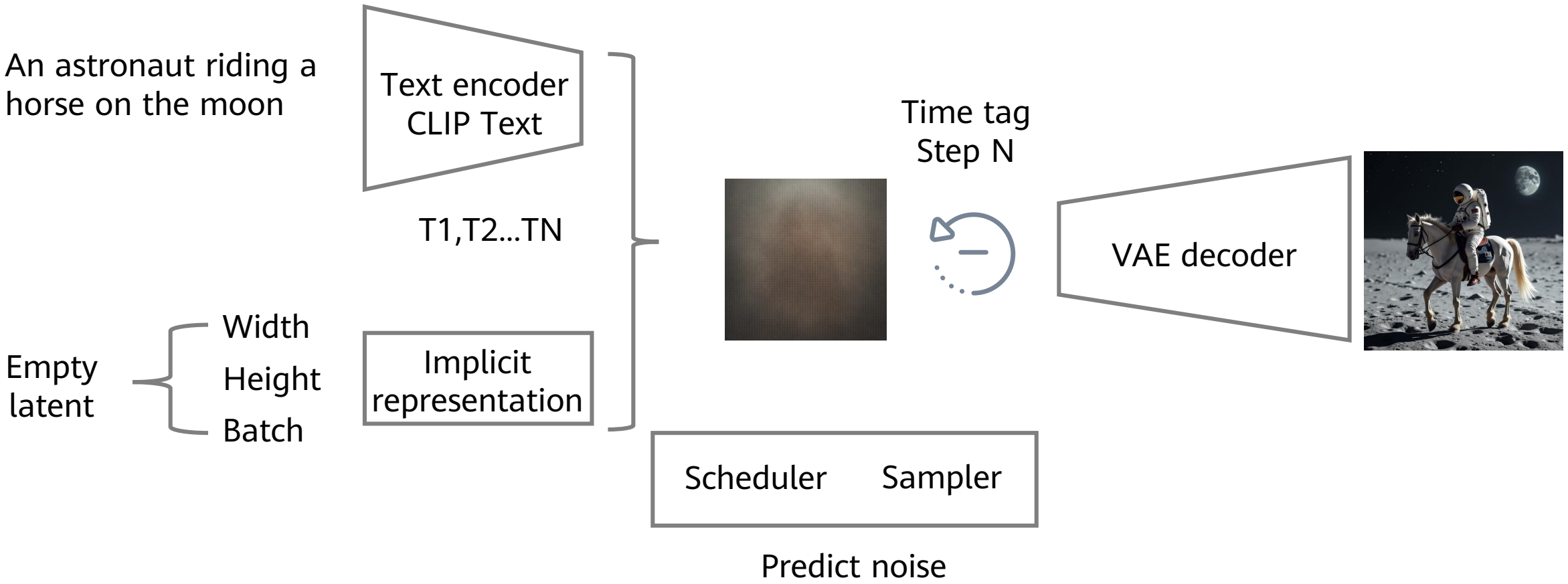
Stable Diffusion Inference (Text-to-Image)



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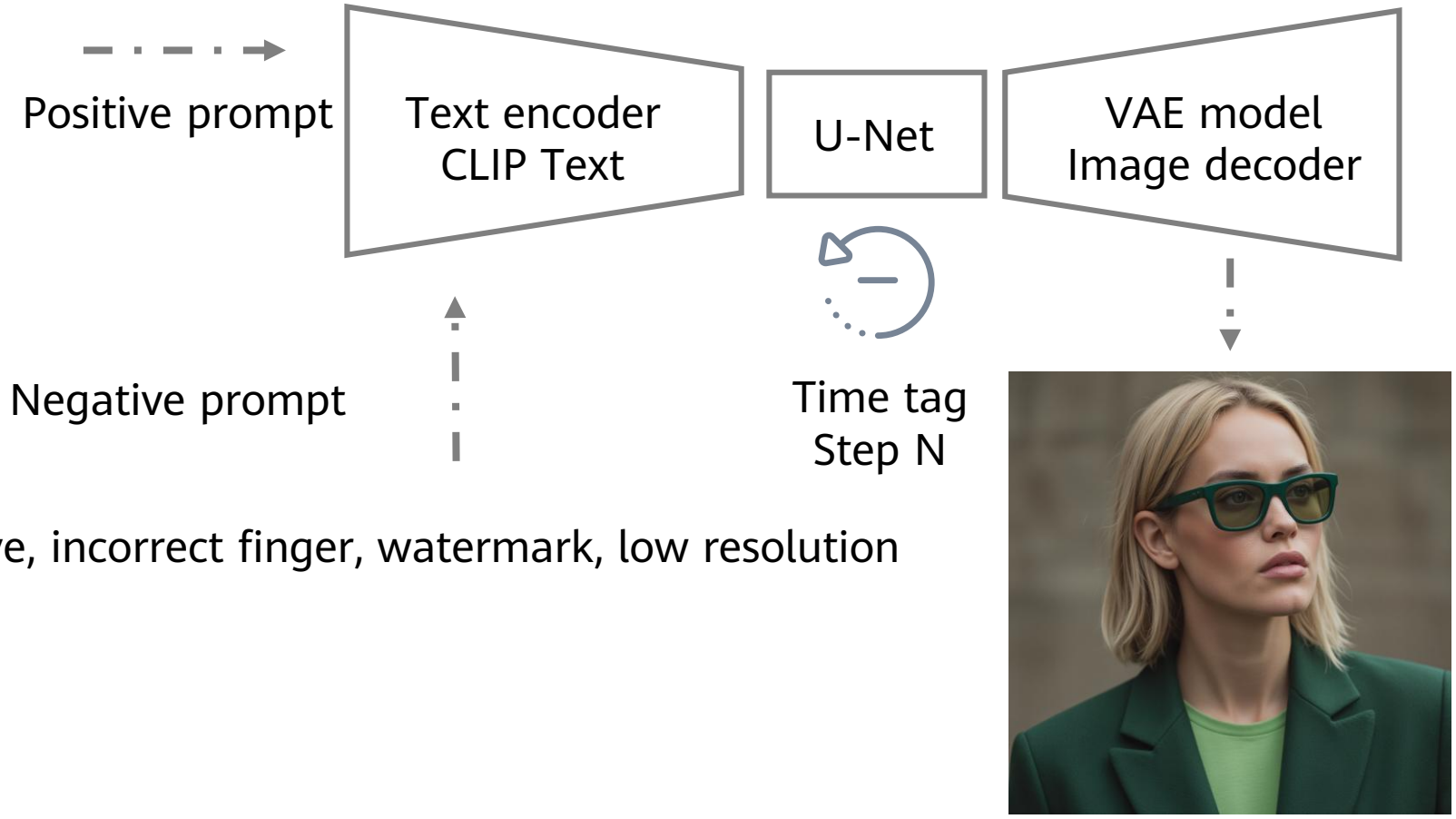
Text-to-Image

- Stable Diffusion can generate images based on text.

A woman wearing green sunglasses and a green blazer is looking towards the right side of the frame.



Incorrect perspective, incorrect finger, watermark, low resolution

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Image-to-Image

- Stable Diffusion can generate images based on text and images.

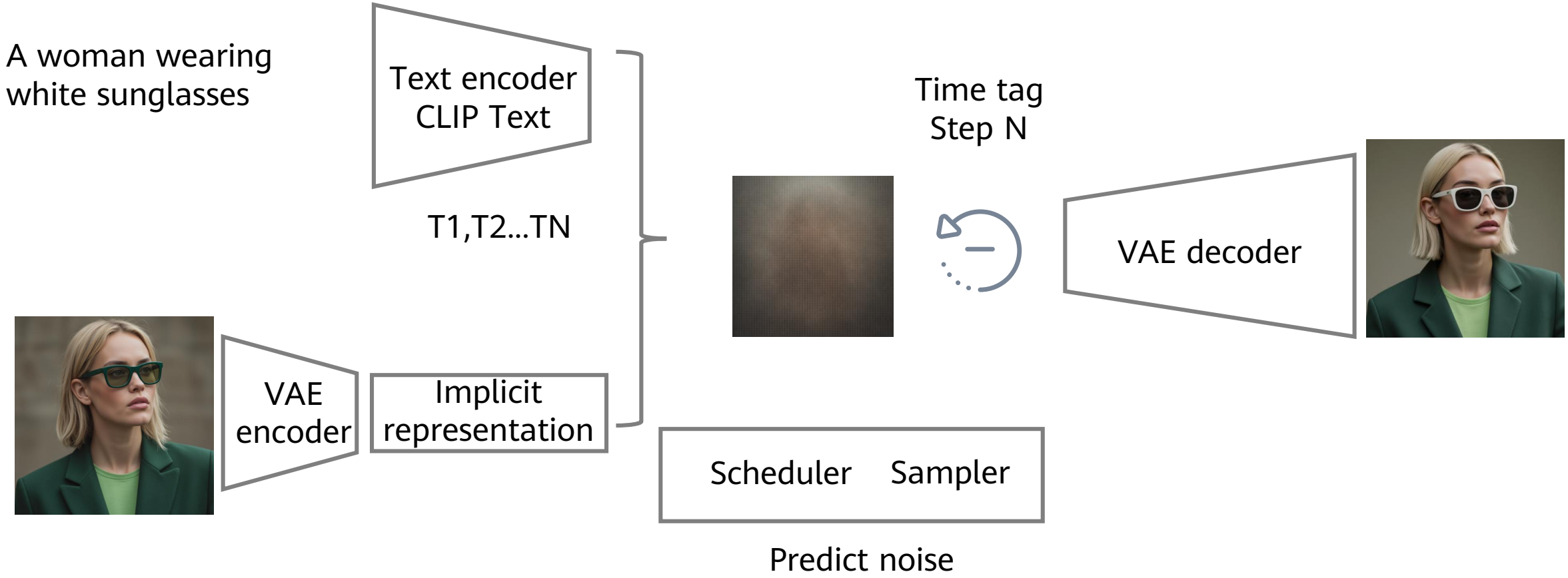

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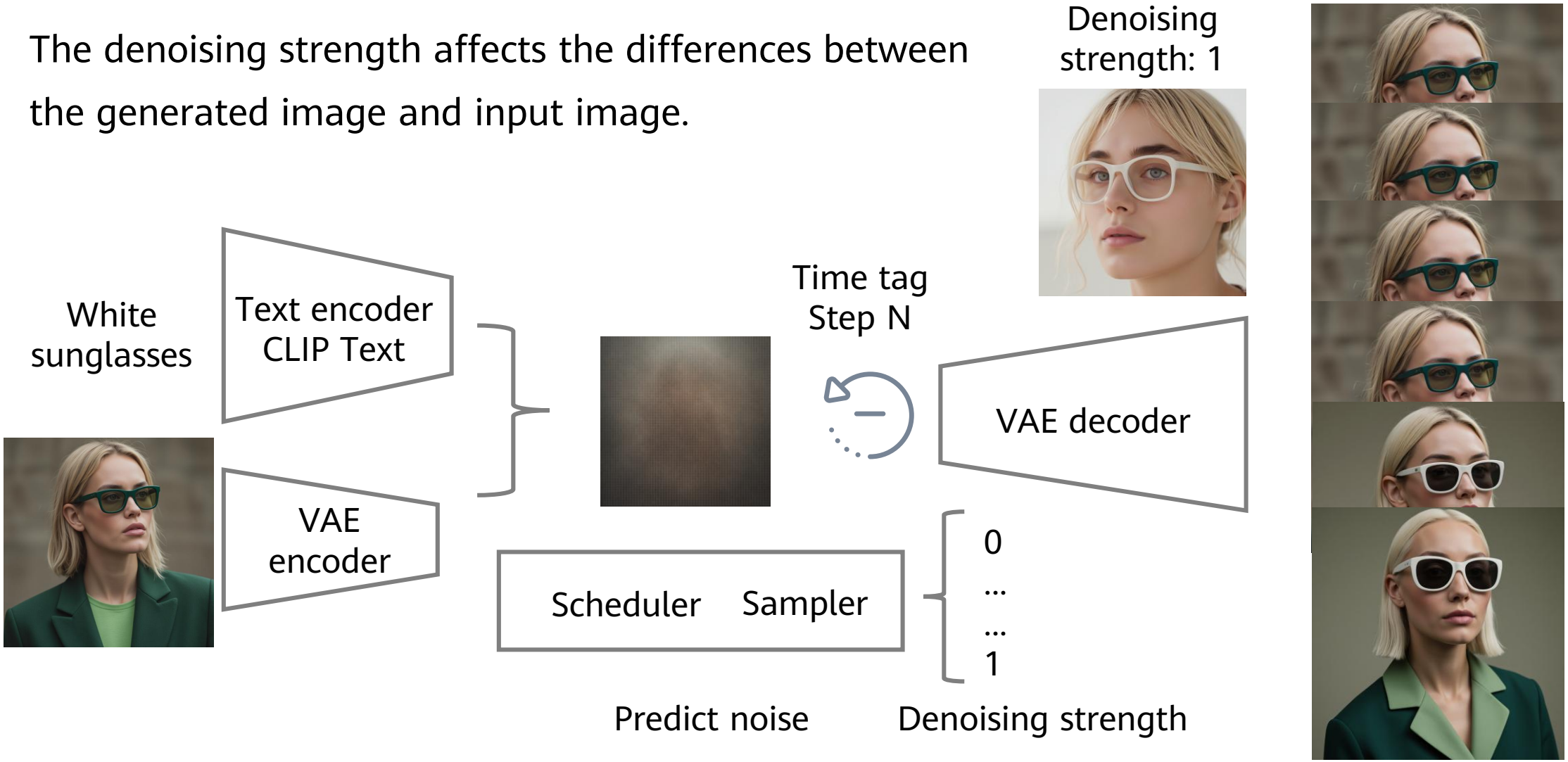
Image-to-Image Denoising Strength

- The denoising strength affects the differences between the generated image and input image.



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Summary

- Basics of diffusion models
- Structure of the Stable Diffusion model
- Basic applications of the Stable Diffusion model

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Quiz

1. Which of the following are the key components of Stable Diffusion? ( )

   A. CLIP

   B. U-Net

   C. VAE

   D. Negative Prompt

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Recommendations

- Huawei Cloud websites
  - Huawei Cloud official website: https://www.huaweicloud.com/intl/en-us/
  - Huawei Cloud Developer Institute: https://edu.huaweicloud.com/intl/en-us/

Huawei Cloud Developer Institute

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